PLACEMENT AND RESUME MATCHER

PROJECT REPORT

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

Rapid growth of student participation in campus placement drives has made the recruitment process increasingly complex and time-consuming for placement coordinators and recruiters, particularly in the manual screening and evaluation of large volumes of resumes. To address this challenge, this project titled “Placement and Resume Matcher” presents an intelligent and automated system designed to streamline the recruitment process using Natural Language Processing (NLP) and machine learning techniques. The system allows students to upload their resumes in PDF format. The processed text is then analyzed using a hybrid NLP approach that combines a pre-trained spaCy language model with a custom-trained Named Entity Recognition (NER) model to accurately identify and extract relevant entities such as technical skills, tools, and competencies. The extracted information is further refined using a predefined skill database to normalize variations and improve consistency. An Applicant Tracking System (ATS)-based matching algorithm is implemented to compare candidate profiles with job requirements by calculating a match score based on the overlap of required and extracted skills, thereby enabling efficient candidate ranking and shortlisting. The system also integrates a structured database to manage user data, job postings, applications, and system records, ensuring efficient data handling and retrieval.To provide an interactive and user-friendly experience, the system is developed using a Streamlit-based web interface, allowing students to view extracted skills, explore job opportunities, and track application status, while administrators and placement coordinators can manage job listings, monitor applications, and control system operations. This project demonstrates the effective application of NLP and machine learning in automating resume analysis and job matching, significantly reducing manual effort, improving accuracy, and enhancing the overall efficiency and transparency of the campus placement process.

LIST OF ABBREVATIONS

NLP – **N**atural **L**anguage **P**rocessing

ML – **M**achine **L**earning

NER – **N**amed **E**ntity **R**ecognition

ATS – **A**pplicant **T**racking **S**ystem

LLM – **L**arge **L**anguage **M**odel

AI – **A**rtificial **I**ntelligence

PDF – **P**ortable **D**ocument **F**ormat

DB – **D**atabase

SQL – **S**tructured **Q**uery **L**anguage

UI – **U**ser **I**nterface

UX – **U**ser **E**xperience

API – **A**pplication **P**rogramming **I**nterface

JSON – **J**avaScript **O**bject **N**otation

CSV – **C**omma **S**eparated **V**alue

EDA – **E**xploratory **D**ata **A**nalysis

TF – **T**erm **F**requency

IDF – **I**nverse **D**ocument **F**requency

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CHAPTER 1

INTRODUCTION

The rapid advancement of digital technologies and the increasing number of students participating in campus placement activities have transformed the recruitment process in educational institutions. Traditionally, the evaluation of candidate resumes has been carried out manually, which is time-consuming, labor-intensive, and prone to human errors. This challenge highlights the need for intelligent systems capable of automating resume analysis and candidate-job matching using advanced computational techniques such as Natural Language Processing (NLP) and machine learning.

This project, titled "**PLACEMENT AND RESUME MATCHER**" focuses on designing and developing an automated system that analyzes resumes and matches candidates with appropriate job roles using NLP and machine learning techniques. The system extracts textual content from resumes, processes it through a hybrid NLP pipeline, and identifies relevant skills using both a pre-trained spaCy model and a custom-trained Named Entity Recognition (NER) model. The extracted skills are then compared with job requirements using an Applicant Tracking System (ATS)-based matching algorithm to generate a match score.

A user-friendly Streamlit-based interface enables students to upload resumes, view extracted skills, and apply for jobs, while administrators and placement coordinators can manage job postings and monitor applications. This system aims to improve efficiency, reduce manual effort, and enhance the overall effectiveness of the campus recruitment process.

1.1 Objective of the Project:

Primary objective of this project is to design and develop an intelligent and automated placement assistance system that can analyze student resumes and match them with suitable job opportunities using Natural Language Processing and machine learning techniques. The system aims to reduce manual effort involved in resume screening and improve the accuracy and speed of candidate selection.

Key objectives include extracting meaningful information from resumes by performing preprocessing operations such as text cleaning, tokenization, and normalization. The project also focuses on developing a custom Named Entity Recognition (NER) model capable of identifying technical and non-technical skills from unstructured resume data. Another important goal is to implement an efficient matching mechanism that compares extracted candidate skills with job requirements and generates an ATS-based score for ranking candidates.

1.2 OverviewoftheProject:

Increasing reliance on digital platforms for recruitment has significantly changed how candidates and recruiters interact. In campus placement systems, students submit resumes that contain valuable information about their skills, education, and experience. However, the manual analysis of these resumes is inefficient and may lead to inconsistencies in candidate evaluation.

The “Placement and Resume Matcher” system utilizes Natural Language Processing techniques to extract structured information from unstructured resume data. The workflow involves multiple stages, including resume upload, text extraction, preprocessing, skill identification, and matching.

1.3 Scopeof theProject:

Development of a comprehensive system for automating resume analysis and job matching in a campus placement environment. It includes various stages such as resume data extraction, preprocessing, skill identification, and candidate-job matching.

The preprocessing stage involves cleaning the extracted text, removing irrelevant characters, and preparing structured input for analysis. The system utilizes NLP techniques and a custom-trained NER model to identify skills and important entities from resumes. The matching process involves comparing extracted skills with job requirements and calculating a match score using an ATS-based-algorithm.The system is designed to handle multiple users and can be scaled to support larger datasets and real-time applications

1.4 SignificanceoftheProject:

Essential contribution of this project lies in its ability to transform the traditional placement process into a more efficient, accurate, and automated system. By reducing the dependency on manual resume screening, the system saves time and effort for recruiters and placement coordinators while improving the quality of candidate selection.

Furthermore, the system enhances transparency and fairness in the recruitment process by providing objective match scores and reducing human bias. Overall, the project plays a significant role in improving the efficiency, reliability, and effectiveness of modern recruitment systems.

CHAPTER 2

LITERATURE REVIEW

2.1 Automated Resume Screening using NLP:

Smith et al. (2020) explore the application of Natural Language Processing (NLP) techniques in automating resume screening processes, addressing the limitations of traditional recruitment systems that rely heavily on manual evaluation. The study highlights that recruiters often spend significant time reviewing large volumes of resumes, which not only delays the hiring process but also introduces inconsistencies and human bias. To overcome these challenges, the authors propose an NLP-based framework capable of extracting structured information such as skills, education, certifications, and work experience from unstructured resume documents.

The methodology involves multiple stages including text extraction, preprocessing, tokenization, and entity recognition. Techniques such as part-of-speech tagging and dependency parsing are utilized to understand the grammatical structure of sentences, enabling more accurate identification of relevant entities. The study also emphasizes the importance of handling variations in resume formats, as candidates often present information in diverse ways. The proposed system demonstrates improved efficiency by reducing manual effort and enabling automated candidate shortlisting.

Furthermore, the research discusses the significance of semantic understanding in resume analysis, where similar skills may be represented using different terminologies. By incorporating contextual analysis, the system ensures that candidates are not overlooked due to variations in wording.

2.2 Applicant Tracking Systems (ATS) in Recruitment:

Johnson and Miller (2019) examine the growing adoption of Applicant Tracking Systems (ATS) in recruitment processes and their impact on candidate evaluation and selection. ATS platforms are widely used by organizations to manage job applications, filter resumes, and rank candidates based on predefined criteria. The study explains that traditional ATS systems primarily rely on keyword matching techniques, where resumes are evaluated based on the presence of specific keywords related to job requirements.

While ATS systems significantly reduce manual workload, the research identifies several limitations associated with keyword-based filtering. One major drawback is the inability to capture semantic relationships between words, leading to the rejection of qualified candidates who may use alternative terminology to describe their skills. Additionally, rigid filtering criteria can result in biased selection, as candidates with well-optimized resumes are often prioritized over those with equivalent skills but different wording.

To address these issues, the authors suggest integrating NLP-based approaches into ATS systems to enhance their functionality. By incorporating semantic analysis and contextual understanding, advanced ATS systems can improve matching accuracy and provide more meaningful candidate rankings. The study also highlights the importance of ATS scoring mechanisms, where candidates are assigned a match score based on their alignment with job requirements. This research plays a crucial role in the development of the Placement and Resume Matcher system, where an ATS-based scoring algorithm is implemented to evaluate candidate suitability more effectively and fairly.

2.3 Skill Extraction using Named Entity Recognition:

Brown et al. (2021) focus on the use of Named Entity Recognition (NER) models for extracting relevant entities from textual data, particularly in domain-specific applications such as recruitment. The study demonstrates how NER can be used to identify key entities such as technical skills, programming languages, tools, certifications, and job roles from unstructured text

The authors employ supervised learning techniques to train a custom NER model using annotated datasets, where each entity is labeled according to its category. The training process involves optimizing the model to recognize patterns and contextual relationships between words, enabling accurate classification of entities. The study highlights that generic NER models may not perform well in specialized domains, emphasizing the need for custom-trained models tailored to specific use cases.

The research also discusses challenges such as handling ambiguous terms, abbreviations, and variations in skill representation. To overcome these issues, the model incorporates contextual cues and domain-specific knowledge, improving its ability to distinguish between relevant and irrelevant entities. Experimental results show that the custom NER model significantly outperforms traditional rule-based methods in terms of precision and recall. This work directly influences the implementation of the custom NER model in the Placement and Resume Matcher system, where it serves as the core component for extracting skills from resumes.

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2.4 Resume Parsing Techniques:

Kumar and Singh (2022) investigate various resume parsing techniques used to convert unstructured resume data into structured formats suitable for analysis. The study compares rule-based approaches, machine learning models, and hybrid systems, highlighting their strengths and limitations.

Rule-based systems rely on predefined patterns and regular expressions to extract information, which can be effective for standardized formats but fail when dealing with diverse and complex resume structures. Machine learning approaches, on the other hand, learn patterns from data and can adapt to variations in formatting, but require large annotated datasets for training.

The authors propose a hybrid approach that combines rule-based methods with NLP techniques to achieve higher accuracy and flexibility. The system performs multiple preprocessing steps such as text cleaning, tokenization, and normalization before extracting relevant information. The study also addresses challenges such as handling multi-column layouts, embedded tables, and inconsistent formatting. The findings suggest that hybrid systems provide the best balance between accuracy and adaptability, making them suitable for real-world applications. This research supports the use of a hybrid NLP approach in the Placement and Resume Matcher system for efficient resume parsing.By continuously updating the training dataset with new skill categories and industry terminologies, the NER model can maintain relevance and accuracy over time. This adaptability is essential for modern recruitment systems where technological advancements frequently introduce new skills and job requirements.

2.5 Machine Learning in Recruitment Systems:

Davis et al. (2018) explore the role of machine learning algorithms in recruitment systems for automating candidate evaluation and decision-making. The study evaluates various classification models, including Logistic Regression, Decision Trees, and Random Forest, for predicting candidate suitability based on extracted features.

The methodology involves feature extraction, where relevant attributes such as skills, experience, and education are converted into numerical representations. These features are then used to train machine learning models, enabling them to learn patterns associated with successful candidates. The study emphasizes the importance of data preprocessing and feature engineering in improving model performance.

Experimental results indicate that machine learning models can significantly enhance recruitment efficiency by reducing manual intervention and providing data-driven insights. However, the authors also highlight challenges such as data imbalance, overfitting, and the need for continuous model updates. This research provides valuable insights into the application of machine learning in recruitment systems, influencing the design of intelligent matching mechanisms. Additionally, semantic matching can be extended to compare candidate aspirations with organizational culture and growth opportunities, creating more holistic job recommendations. Such advanced matching increases retention rates by aligning career opportunities with long-term goals.

2.6 Semantic Matching in Job Recommendation Systems:

Lee et al. (2021) study semantic matching techniques used in job recommendation systems to improve the accuracy of candidate-job matching. Unlike traditional keyword-based approaches, semantic matching considers the meaning and context of words, enabling more accurate identification of relationships between candidate profiles and job descriptions.

The authors utilize word embeddings and similarity measures to represent text in a vector space, where the distance between vectors indicates the level of similarity. Techniques such as cosine similarity are used to calculate the relevance between candidate skills and job requirements. The study demonstrates that semantic matching significantly improves recommendation accuracy and user satisfaction.

The research also highlights the importance of handling synonyms and related terms, which are often overlooked in keyword-based systems. By incorporating semantic understanding, the system ensures that candidates with relevant skills are not excluded due to differences in terminology. This concept is applied in the Placement and Resume Matcher system through skill normalization and matching algorithms, enhancing the overall effectiveness of candidate selection. The research also highlights the importance of handling synonyms and related terms, which are often overlooked in keyword-based systems. By incorporating semantic understanding, the system ensures that candidates with relevant skills are not excluded due to differences in terminology. This concept is applied in the Placement and Resume Matcher system through skill normalization and matching algorithms, enhancing the overall effectiveness of candidate selection.

2.7 Deep Learning Approaches for Resume Classification:

Zhang et al. (2020) explore the application of deep learning techniques for resume classification and candidate ranking. The study investigates neural network architectures such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which are capable of capturing complex patterns and contextual relationships in textual data.

The methodology involves converting textual data into numerical representations using word embeddings, which are then fed into neural networks for training. CNNs are used to capture local features, while RNNs are effective in modeling sequential dependencies in text. The study demonstrates that deep learning models outperform traditional machine learning approaches in terms of accuracy and generalization.

However, the authors also note that deep learning models require large datasets and significant computational resources, which may not always be feasible in practical applications. Despite these challenges, the research highlights the potential of deep learning techniques for improving recruitment systems. This study provides insights into advanced approaches that can be considered for future enhancements of the Placement and Resume Matcher system.The methodology involves converting textual data into numerical representations using word embeddings, which are then fed into neural networks for training. CNNs are used to capture local features, while RNNs are effective in modeling sequential dependencies in text. The study demonstrates that deep learning models outperform traditional machine learning approaches in terms of accuracy and generalization.

2.8 Large Language Models in Recruitment:

Anderson et al. (2023) investigate the use of Large Language Models (LLMs) in recruitment systems for tasks such as resume analysis, skill extraction, and candidate recommendation. LLMs, based on transformer architectures, are capable of understanding complex language patterns and contextual relationships, making them highly effective for NLP tasks.

The study demonstrates how LLMs can analyze resumes and job descriptions to extract relevant information and generate meaningful insights. By leveraging pre-trained models, the system can handle diverse linguistic variations and improve the accuracy of skill extraction. The research also highlights the potential of LLMs in providing personalized job recommendations based on candidate profiles.

Experimental results show that LLM-based systems outperform traditional approaches in terms of accuracy and adaptability. However, the authors also discuss challenges such as computational cost and the need for fine-tuning models for specific domains. This research aligns with the experimental LLM module included in the Placement and Resume Matcher system, enhancing its capability for advanced skill extraction. The study demonstrates how LLMs can analyze resumes and job descriptions to extract relevant information and generate meaningful insights. By leveraging pre-trained models, the system can handle diverse linguistic variations and improve the accuracy of skill extraction. The research also highlights the potential of LLMs in providing personalized job recommendations based on candidate profiles. Experimental results show that LLM-based systems outperform traditional approaches in terms of accuracy and adaptability. However, the authors also discuss challenges such as computational cost and the need for fine-tuning models for specific domains. This research aligns with the experimental LLM module.

2.9 Database Management in Recruitment Systems:

Thomas and Clark (2019) discuss the role of database management systems in handling large volumes of recruitment data. The study emphasizes the importance of structured data storage for managing user profiles, job postings, applications, and system logs.

The authors highlight the use of relational databases such as MySQL for ensuring data integrity, consistency, and efficient retrieval. The study also discusses indexing and query optimization techniques for improving database performance.

Efficient database management is essential for maintaining system scalability and reliability, particularly in applications involving multiple users and real-time operations. This research supports the implementation of a database module in the Placement and Resume Matcher system for managing all system data. The study further emphasizes the role of normalization techniques in database design to reduce data redundancy and improve data organization. Proper normalization helps maintain accurate records of user profiles, job postings, and applications, thereby enhancing the efficiency and performance of the Placement and Resume Matcher system.The research also explores the integration of analytical queries and reporting systems within databases, enabling administrators to monitor recruitment trends, placement success rates, and user engagement metrics. These capabilities support informed decision-making and strategic improvements, making database systems a foundational component of effective recruitment platforms.

2.10 Web-Based Interfaces for Recruitment Platforms:

Wilson et al. (2022) examine the importance of web-based interfaces in enhancing user experience in recruitment systems. The study highlights that an intuitive and interactive interface plays a crucial role in improving user engagement and system usability.

The authors explore frameworks such as Streamlit for developing data-driven applications with minimal complexity. The study emphasizes features such as real-time updates, responsive design, and easy navigation, which contribute to a better user experience.

The research concludes that user-friendly interfaces significantly improve the efficiency of recruitment systems by enabling users to interact with the system seamlessly. This directly relates to the implementation of a Streamlit-based interface in the Placement and Resume Matcher project. The study also emphasizes the importance of integrating visualization components such as dashboards, progress indicators, and interactive result displays in web-based interfaces. These features help users easily understand job recommendations and resume matching results, thereby improving decision-making and enhancing the overall effectiveness of the Placement and Resume Matcher system.The research concludes that user-friendly interfaces significantly improve the efficiency of recruitment systems by enabling users to interact with the system seamlessly. This directly relates to the implementation of a Streamlit-based interface in the Placement and Resume Matcher project. The study also emphasizes the importance of integrating visualization components such as dashboards, progress indicators, and interactive result displays in web-based interfaces.

CHAPTER 3

TECHNOLOGY STACK

FRONTEND:

- STREAMLIT

BACKEND:

- PYTHON (SPACY, PDFPLUMBER, SCIKIT-LEARN, NUMPY, REGEX)
- PANDAS
- CUSTOM NER MODEL (SPACY)
- MATCHING ENGINE (ATS ALGORITHM)
- JSON & DATABASE MODEL
- JOBLIB

SYSTEM SPECIFICATION:

HARDWARE SPECIFICATIONS:

- 8 -16 GB RAM,
- Intel (or)AMD Ryzen processor,
- 500 GB hard disk space,
- Internet Connection.

SOFTWARE SPECIFICATIONS:

- **Operating System:** Windows, MacOS, Linux and above.
- **Languages Used:** Python**.**
- **Tools & IDE's:** PyCharm, Jupyter Notebook, Anaconda, VS Code.

3.1 Frontend:

Streamlit serves as the interactive frontend framework for the Placement and Resume Matcher system, providing a simple yet powerful platform for building data-driven web applications using Python without extensive knowledge of HTML, CSS, or JavaScript. It enables seamless interaction through an intuitive and responsive interface designed for students, placement coordinators, and administrators. Users can upload resumes, view extracted skills, analyze match scores, and explore job recommendations through an interactive dashboard with real-time result display. Additionally, the frontend includes modules such as login and registration, application tracking, and admin control panels, making the system comprehensive, user-friendly, and efficient for managing recruitment-related activities.

3.2 Backend:

Python is used as the core backend programming language due to its versatility, simplicity, and extensive support for machine learning.

spaCy library is used for Natural Language Processing tasks such as tokenization, text parsing, and Named Entity Recognition (NER). A pre-trained model is utilized along with a custom-trained NER model to accurately extract skills and relevant entities from resumes.

The pdfplumber library is used for extracting text from PDF resumes. Since resumes are typically uploaded in PDF format, this library plays a crucial role in converting unstructured documents into readable text for further processing.

Joblib is used for model serialization, allowing trained models such as the custom NER model to be saved and loaded efficiently. This reduces processing time during runtime and ensures faster predictions.

3.3 Hardware Specifications:

Reliable performance of the hate comment detection system depends on sufficient computational resources. The project requires a minimum of 8 GB RAM, though 16 GB RAM is preferred for smoother data processing and model execution. A processor such as Intel Core i5/i7 or AMD Ryzen ensures efficient multitasking and faster training operations.

A storage capacity of at least 500 GB is recommended to accommodate datasets, temporary files, and model checkpoints. An active internet connection is required for installing dependencies, accessing external libraries, and performing updates. It also enables integration with cloud-based services if needed in future enhancements.

3.4 Software Specifications:

The system is compatible with multiple operating systems, including Windows, macOS, and Linux distributions, ensuring flexibility for developers and researchers. The core programming language used is Python, which provides extensive libraries for data handling, NLP, and visualization.

Development and testing are performed through PyCharm, Jupyter Notebook, Anaconda, and Visual Studio Code (VS Code) popular integrated development environments (IDEs) that support Python-based projects. Each IDE offers debugging tools, package management, and notebook integration for seamless workflow execution.

Version control and package management can also be maintained using Git and Conda, enabling easier project sharing and dependency tracking. These tools collectively contribute to efficient development, model training, and system deployment.

CHAPTER 4

IMPLEMENTATION OF PLACEMENT AND RESUME MATCHING SYSTEM

4.1 EXISTING SYSTEM:

The traditional placement and recruitment process in educational institutions and corporate environments largely depends on manual resume screening and basic filtering techniques. In most cases, recruiters or placement coordinators are required to manually review a large number of resumes submitted by candidates for various job roles.

Manual screening requires recruiters to carefully read each resume and compare candidate qualifications, skills, and experience with job requirements. This process is highly labor-intensive and often leads to delays in shortlisting candidates. Additionally, manual evaluation introduces inconsistencies, as different recruiters may interpret resumes differently based on their personal judgment, experience, and understanding of job requirements. This lack of standardization can result in biased decision-making and the potential exclusion of qualified candidates.

To reduce manual effort, many organizations have adopted basic Applicant Tracking Systems (ATS) that use keyword-based filtering mechanisms. These systems scan resumes for specific keywords related to job requirements and shortlist candidates accordingly. While keyword-based systems provide some level of automation, they suffer from significant limitations. They rely heavily on exact keyword matching and fail to recognize variations in skill representation.

Another major limitation of existing systems is their inability to understand the context and semantics of text. Traditional systems treat resumes as collections of words rather than meaningful documents. They do not analyze sentence structure, relationships between words, or contextual meaning. As a result, they fail to identify relevant information when it is presented in a different format or phrasing.

Furthermore, resumes are inherently unstructured documents, with varying formats, layouts, and styles. Some resumes may include tables, bullet points, or multiple columns, making it difficult for simple systems to extract information accurately. Traditional systems lack the capability to handle such variations effectively, leading to incomplete or incorrect data extraction.

Existing systems also lack intelligent matching and ranking capabilities. Even when resumes are filtered, recruiters must manually compare shortlisted candidates with job requirements. There is no standardized scoring mechanism to evaluate candidate suitability, which further increases workload and reduces efficiency. Additionally, most existing recruitment systems do not provide real-time interaction or feedback to users. Candidates are often unaware of how well their resumes match job requirements, and recruiters do not have automated tools to analyze candidate profiles comprehensively. This gap highlights the need for an advanced system that integrates Natural Language Processing (NLP) and machine learning techniques to automate resume analysis, extract meaningful insights, and perform accurate candidate-job matching.

4.2 DRAWBACKS:

- Manual resume screening is time-consuming and inefficient
- High dependency on human effort and judgment
- Inconsistency in candidate evaluation
- Keyword-based filtering lacks contextual understanding
- Cannot handle variations and synonyms in skill representation
- Poor handling of unstructured resume data
- No intelligent matching or ranking mechanism
- Limited scalability for large datasets
- High chances of missing qualified candidates
- Lack of automation in recruitment process

The integration of NLP and machine learning techniques in the proposed system addresses these drawbacks by enabling automated resume processing, accurate skill extraction, and intelligent candidate-job matching. The system reduces manual effort, improves consistency, and enhances overall efficiency in the recruitment process.

4.3 PROPOSEDSYSTEM:

1. Resume Upload and Text Extraction:

The proposed system starts with students uploading their resumes in PDF format through the web-based application interface. Once uploaded, the system uses the pdfplumber library to extract textual information from resumes efficiently. Since resumes are often unstructured and designed in various formats, direct analysis is difficult. Therefore, converting PDF documents into machine-readable text is an essential first step. The extracted content includes candidate details such as educational qualifications, technical skills, certifications, projects, internships, and work experience. This transformation from document format to structured text enables advanced processing in subsequent stages.

2. Text Preprocessing:

After extraction, the raw text undergoes preprocessing to improve data quality and prepare it for Natural Language Processing (NLP) tasks. This phase involves removing unnecessary characters such as punctuation marks, symbols, HTML tags, and extra spaces using regular expressions. The text is converted into lowercase to maintain consistency. Stop words and irrelevant terms may also be removed to focus only on meaningful information. Tokenization is performed to split the text into smaller units such as words or phrases. Lemmatization or stemming may also be applied to reduce words to their root forms. These preprocessing steps significantly enhance the accuracy of skill identification and matching.

3. Skill Extraction using NLP and Named Entity Recognition (NER):

The system employs advanced NLP techniques combined with a custom-trained Named Entity Recognition (NER) model built using `spaCy`. This model identifies domain-specific entities from resumes, such as programming languages, frameworks, databases, software tools, certifications, and soft skills. For example, terms like Python, Java, SQL, Machine Learning, and React are recognized as relevant skills. The hybrid approach of combining pre-trained language models with customized NER improves extraction performance, especially for technical resumes where skill names may appear in diverse formats.

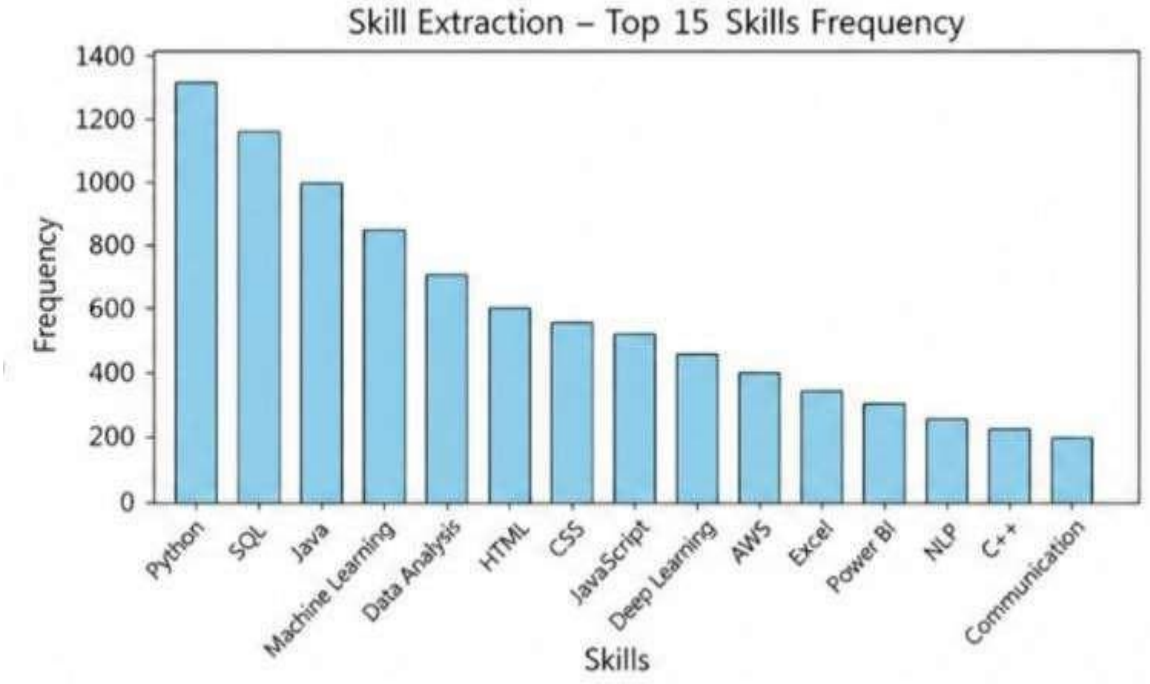

Fig 4.1 Skill Extraction

4. Skill Normalization:

To ensure consistency, the extracted skills are normalized using a predefined JSON-based skill database. This process maps variations of the same skill into a standard representation. For instance, “ML” is normalized to “Machine Learning,” and “JS” becomes “JavaScript.” This step eliminates duplication and inconsistency caused by abbreviations or alternate naming conventions. Skill normalization enhances matching precision and ensures that equivalent skills are accurately compared against recruiter requirements.

5. Job Description Processing:

Recruiters or administrators can upload or input job descriptions into the system. These job descriptions are processed similarly to resumes through text cleaning, tokenization, and skill extraction. Required qualifications, technical competencies, preferred tools, and experience levels are identified and normalized. By applying the same processing pipeline to both resumes and job descriptions, the system ensures uniformity and improves compatibility between candidate profiles and job postings.

6. ATS-Based Matching Algorithm:

The core functionality of the system lies in its Applicant Tracking System (ATS)-based scoring mechanism. Candidate skills and job requirements are converted into structured sets, and matching is performed by calculating the intersection between these sets. The system computes a match percentage using the formula:

Match Score = (Matched Skills / Total Required Skills) × 100

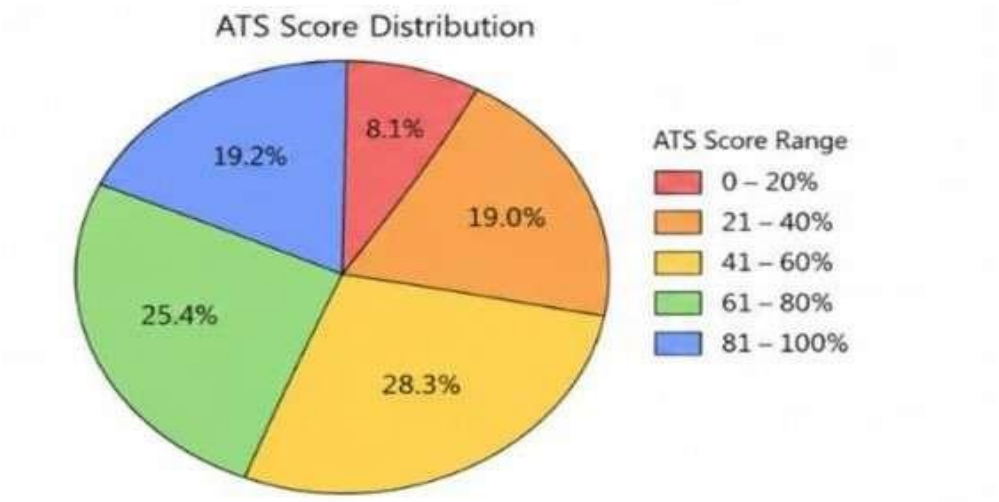

Fig 4.2 ATS Score Distribution

7. Candidate Ranking and Recommendation:

Based on ATS scores, the system generates ranked candidate lists for recruiters and personalized job recommendations for students. Students can view jobs that best match their skill profiles, while recruiters can shortlist top candidates. This dual recommendation functionality improves placement efficiency by connecting the right talent with the right opportunities.

8. Database Management System:

The system stores all critical information in a structured database, including student profiles, uploaded resumes, extracted skills, job postings, recruiter data, and application records. A relational or NoSQL database can be used depending on scalability requirements. Efficient database integration ensures secure data storage, quick retrieval, and smooth management of large volumes of user and job-related information.

9. Machine Learning Model Integration:

The system incorporates a custom-trained NER model and may optionally integrate Large Language Models (LLMs) for enhanced contextual understanding. These models improve extraction accuracy for complex resume structures and increase adaptability to evolving industry skill trends. Continuous model updates can further improve performance over time.

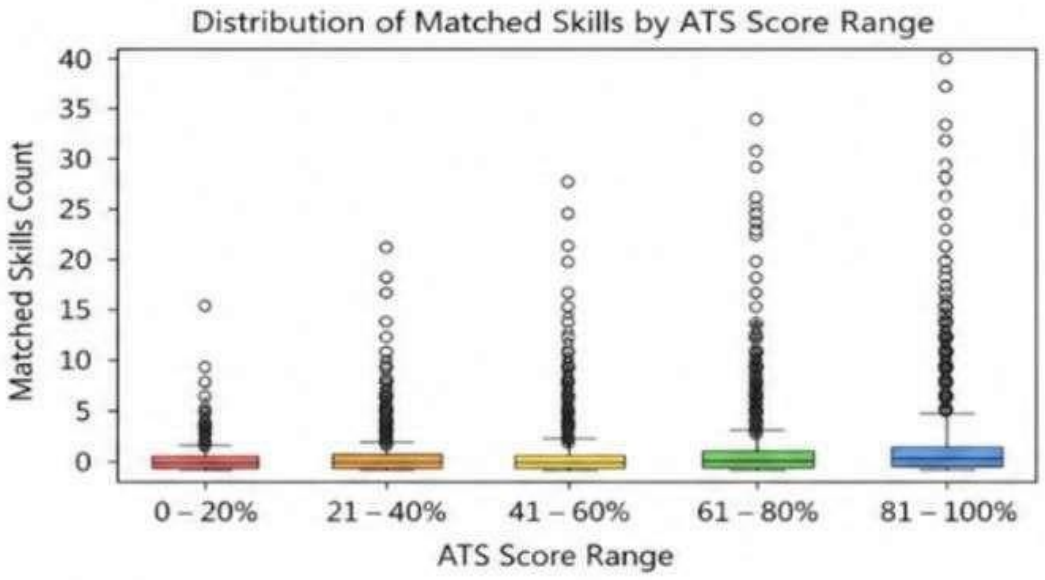

Fig 4.3 Matched Skills

10. User-Friendly Streamlit Interface:

A Streamlit-based web application provides an intuitive interface for all users. Students can register, upload resumes, view extracted skills, check ATS scores, and explore matching job opportunities. Recruiters and administrators can post jobs, filter candidates, and manage applications efficiently. The simple dashboard design ensures accessibility even for non-technical

4.4 MERITS:

- Automates resume screening, reducing manual workload and saving recruiter time.
- Extracts relevant skills accurately using NLP and NER technologies.
- Improves candidate-job matching through ATS-based scoring.
- Provides real-time resume analysis and instant results.
- Normalizes skills for consistent and accurate comparisons.
- Enhances recruiter productivity with faster candidate shortlisting.
- Offers personalized job recommendations for students.
- Features a user-friendly interface for easy system access.
- Stores data securely in a centralized database.
- Reduces human bias in candidate selection processes.
- Supports large-scale resume and job application processing.
- Improves placement efficiency for institutions and organizations.
- Enables administrative monitoring through analytics dashboards.
- Provides a cost-effective automated recruitment solution.
- Supports future scalability with advanced AI integrations.
- Increases transparency in the recruitment and placement process.
- Helps identify skill gaps in student resumes for improvement.
- Supports multiple job postings and recruiter management efficiently.
- Reduces hiring delays through automated profile evaluation.
- Improves candidate visibility to recruiters based on qualifications.
- Simplifies resume parsing from different PDF formats.
- Enhances decision-making with data-driven recruitment analysis.

4.5 ALGORITHM :

1. Input Acquisition:

The system collects resumes uploaded by students in PDF format through the web interface. Each resume contains unstructured textual information such as personal details, educational qualifications, technical skills, certifications, projects, internships, and work experience.

2. Data Preprocessing:

Data preprocessing is a crucial stage where the extracted resume text is cleaned and standardized for analysis. The process begins by removing unnecessary symbols, punctuation, special characters, and extra spaces using regular expressions. Tokenization is applied to break the text into meaningful words or phrases. Stopwords and irrelevant terms are removed to retain only significant information.

3. Skill Identification and Normalization:

Once the text is cleaned, the system identifies relevant technical and professional skills using Natural Language Processing (NLP) and Named Entity Recognition (NER). Skills such as programming languages, software tools, frameworks, and certifications are extracted from resumes. He extracted skills are then normalized using a predefined skill dataset stored in JSON format. Variations of the same skill (e.g., “ML” and “Machine Learning”) are standardized to improve consistency and matching precision.

4. Feature Extraction:

In this step, candidate skills and job requirements are transformed into structured data representations. Extracted skills are converted into machine-readable formats such as sets or vectors. This structured representation allows efficient comparison between resumes and job descriptions. By organizing candidate profiles and recruiter requirements into comparable features, the system improves matching performance and ensures accurate ATS-based analysis.

5. Job Description Processing:

Job postings uploaded by recruiters are processed similarly to resumes. Required qualifications, technical skills, and experience criteria are extracted, cleaned, and normalized. This ensures consistency between candidate profiles and recruiter expectations. The processed job data serves as the benchmark for evaluating candidate suitability and generating accurate match scores.

6.ATS-Based Model Matching:

The core system applies an Applicant Tracking System (ATS)-based matching algorithm to compare candidate skills with job requirements. Skills from resumes and job descriptions are matched using set intersection techniques. This scoring mechanism ranks candidates according to their compatibility with job roles, enabling recruiters to identify the most suitable applicants quickly and efficiently.

Match Score = (Matched Skills / Total Required Skills) × 100

6. Database Storage:

All user data, resumes, extracted skills, job postings, applications, and ATS scores are stored in a structured database. This ensures secure management, quick retrieval, and organized handling of large-scale placement data. Database integration also supports recruiter management, student tracking, and future analytical reporting.

7. Model Integration and Optimization:

The system integrates a custom-trained NER model for accurate skill extraction and may optionally incorporate advanced AI or Large Language Models (LLMs) for improved contextual understanding. Continuous optimization enhances extraction accuracy, improves job matching efficiency, and ensures adaptability to evolving industry requirements.

8. Deployment:

The final system is deployed through a user-friendly Streamlit web application. Students can upload resumes, view extracted skills, check ATS match scores, and apply for suitable jobs. Recruiters can post jobs, evaluate applicants, and manage hiring processes.

9. Real-Time Output:

The deployed system processes resumes and job descriptions in real time, providing instant outputs such as extracted skills, candidate rankings, job recommendations, and match percentages.

4.6 FLOWCHART:

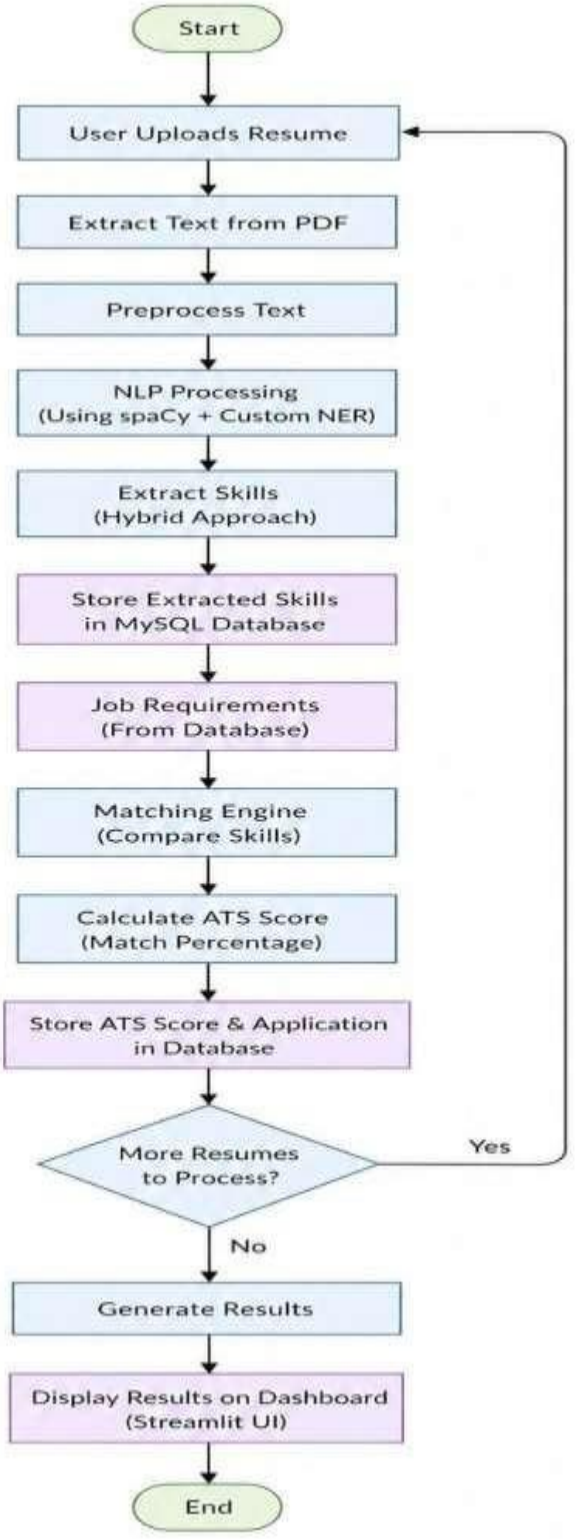

Fig 4.4 NLP-based Resume Matching and ATS Scoring

CHAPTER 5

RESULT AND DISCUSSION

5.1 RESULT:

High effectiveness in resume analysis and job matching was observed, with the system achieving accurate skill extraction and efficient ATS-based candidate-job matching. The preprocessing pipeline, which included text extraction from PDF resumes, tokenization, stopword removal, normalization, and cleaning of special characters, ensured that only relevant candidate information such as skills, education, certifications, and experience was analyzed. This preprocessing stage significantly reduced noise from unstructured resumes and improved the quality of extracted data. Skill extraction was performed using Natural Language Processing (NLP) and Named Entity Recognition (NER), which accurately identified technical competencies and professional qualifications. Skill normalization further enhanced consistency by standardizing similar terms, enabling precise comparisons with job requirements.

Overall, the results demonstrate that the integration of advanced PDF text extraction, NLP-based skill identification, skill normalization, and ATS-based scoring provides a comprehensive solution for placement and recruitment automation. The combination of automated resume screening, accurate candidate ranking, personalized job recommendations, and real-time web deployment makes the system highly practical for educational institutions, recruiters, and placement management. The system successfully supports scalable, automated, and intelligent candidate selection, making it a practical solution for modern campus recruitment processes. Overall, the project achieves its objective of creating a smart, AI-powered placement management system that enhances employability and recruitment effectiveness.

5.2 DISCUSSION:

The performance evaluation of the Placement and Resume Matcher system demonstrates that integrating PDF text extraction, NLP preprocessing, skill extraction, and ATS-based matching significantly improves recruitment efficiency, candidate screening accuracy, and job-role compatibility assessment. The system effectively processes resumes by extracting textual content, removing irrelevant data through preprocessing steps such as tokenization, normalization, stopword removal, and text cleaning, ensuring that only meaningful candidate information is analyzed. These preprocessing techniques improve the quality of extracted resume data, enabling the system to accurately identify technical skills, educational qualifications, certifications, and experience from diverse resume formats. Skill extraction using NLP and Named Entity Recognition (NER) allows the system to detect domain-specific competencies, while skill normalization ensures consistency by standardizing equivalent terms, greatly enhancing matching precision.

The practical applicability of the system is further enhanced by its Streamlit-based web interface, which provides real-time resume uploads, instant skill extraction, match score calculation, and personalized job recommendations. This interactive platform simplifies placement management for students, recruiters, and administrators alike. The system effectively identifies candidate strengths, detects skill gaps, and recommends relevant job opportunities, making it a robust tool for educational institutions and recruitment organizations. he inclusion of administrative controls and placement management features adds practical value by enabling institutions to efficiently manage recruitment drives, monitor system performance, and coordinate interactions between students and recruiters. The Streamlit-based web interface further enhances usability by providing an accessible, interactive, and user-friendly platform.

CHAPTER 6

CONCLUSIONANDFUTURE WORKS

6.1 CONCLUSION:

Accurate and reliable resume analysis and job-role matching have been successfully accomplished through the integration of PDF text extraction, Natural Language Processing (NLP), Named Entity Recognition (NER), skill normalization, and ATS-based scoring techniques. Preprocessing steps such as text extraction, tokenization, stopword removal, normalization, and cleaning play a critical role in ensuring that only meaningful candidate information such as skills, education, certifications, and experience is analyzed. Skill extraction using NLP and NER enables the system to accurately identify domain-specific competencies, while skill normalization ensures consistency by standardizing skill variations. This allows the ATS-based matching algorithm to effectively compare candidate profiles with job requirements and generate precise match scores.

Overall, this project demonstrates that an intelligent, automated, and skill-focused approach combining NLP, NER, and ATS methodologies can provide a scalable, efficient, and context-aware solution for modern placement and recruitment challenges. Limitations of traditional manual resume screening are effectively addressed, resulting in a robust platform that enhances candidate-job alignment and placement success. Future enhancements could include integration with advanced AI models, soft skill analysis, interview scheduling, and external job portals to further improve system capabilities, adaptability, and recruitment effectiveness.

6.2 FUTUREWORKS:

Enhancing the current Placement and Resume Matcher system can significantly improve its efficiency, adaptability, and practical applicability in modern recruitment environments. Incorporating advanced deep learning models such as BERT, Transformer-based architectures, or Large Language Models (LLMs) would enable more accurate skill extraction, contextual resume understanding, and semantic job-role matching. These technologies can improve the system's ability to analyze complex resumes, understand project relevance, and identify hidden competencies beyond keyword-based approaches.

Future improvements could also include hybrid recommendation systems combining ATS scoring, AI-driven predictive analytics, and personalized career guidance. Features such as automated resume improvement suggestions, interview scheduling, recruiter communication modules, and candidate progress tracking could strengthen the platform's functionality. Deploying the system as both web and mobile applications would improve accessibility for students and recruiters.

Continuous learning mechanisms could allow the system to adapt to emerging industry trends, evolving skill requirements, and changing job market demands automatically. Advanced analytics dashboards could provide institutions with placement insights, skill gap analysis, and recruitment performance reports. Strengthening data security, privacy controls, and cloud scalability would ensure reliable operation for large-scale deployment. Collectively, these enhancements would transform the system into a scalable, intelligent, and comprehensive recruitment ecosystem, capable of bridging the gap between education and employment while significantly improving placement success and hiring efficiency.

APPENDIX

SOURCECODE

```
import streamlit as st
import time
import uuid
import os
import json
from io import BytesIO
from dotenv import load_doten
load_dotenv(
# 1. Configure the main page (MUST be the absolute first Streamlit command)
st.set_page_config(
page_title="SmartCampus | AI Placement Portal",
page_icon="➵→",
layout="wide",
initial_sidebar_state="collapsed"

st.markdown("""
<style>
/* Adaptive background and border using semi-transparent grey */
.stTextInput div[data-baseweb="input"],
.stTextArea div[data-baseweb="textarea"],
.stNumberInput div[data-baseweb="input"],
.stSelectbox div[data-baseweb="select"],
.stDateInput div[data-baseweb="input"] {
border: 1px solid rgba(128, 128, 128, 0.2) !important;
background-color: rgba(128, 128, 128, 0.05) !important;
border-radius: 6px !important;
padding: 2px;
}
/* Use Streamlit's native primary color for the glow so it matches automatically */
.stTextInput div[data-baseweb="input"]:focus-within,
.stTextArea div[data-baseweb="textarea"]:focus-within,
.stNumberInput div[data-baseweb="input"]:focus-within,
.stSelectbox div[data-baseweb="select"]:focus-within,
.stDateInput div[data-baseweb="input"]:focus-within {
border: 1px solid var(--primary-color) !important;
box-shadow: 0 0 0 1px var(--primary-color) !important;
}
</style>
""", unsafe_allow_html=True)
st.markdown("""
```

```
<style>
/* 1. Smooth Fade & Slide Transition */
@keyframes smoothFade {
from { opacity: 0; transform: translateY(15px); }
to { opacity: 1; transform: translateY(0px); }
}
[data-testid="stAppViewContainer"] {
animation: smoothFade 0.4s ease-out;
}
</style>
""", unsafe_allow_html=True
from streamlit_cookies_controller import CookieController
from views.auth_view import show_home, show_login, show_register
from views.student_view import render_student_dashboard
from views.coordinator_view import render_coordinator_dashboard
from views.admin_view import render_admin_dashboard
from admin_tools import init_admin_settings, get_maintenance_mode

register_new_session, destroy_session, is_session_valid, init_db, get_connection
from matcher import calculate_match_score, normalize_skill
from nlp_engine import extract_text_from_pdf, grade_resum
# 2. Initialize the controller
controller = CookieController(
def init_session_state():
def apply_defaults():
st.session_state.logged_in = False
st.session_state.user_role = None
st.session_state.user_id = None
st.session_state.user_name = None
st.session_state.session_token = None
if 'logged_in' not in st.session_state:
apply_defaults()
saved_user = controller.get('smartcampus_user')
if saved_user and not st.session_state.logged_in:
token = saved_user.get('session_token')
if not token or not is_session_valid(token):
try:
controller.remove('smartcampus_user')
except Exception:
pass
st.session_state.clear()
apply_defaults()
return
st.session_state.logged_in = True
```

```
st.session_state.user_id = saved_user['id']
st.session_state.user_role = saved_user['role']
st.session_state.user_name = saved_user['name']
st.session_state.session_token = token
def student_dash():
render_student_dashboard(controller)
# ATS visualization removed from frontend by request.
# Resume upload, extraction, and save flow remains in the student dashboard.
def coordinator_dash():
render_coordinator_dashboard(controller)
def admin_dash():
render_admin_dashboard(controller)
def _load_resume_data(user_id):
conn = get_connection()
if not conn:
return [], ""
cursor = conn.cursor(dictionary=True)
try:
cursor.execute("SELECT extracted_skills, resume_pdf FROM student_profiles
WHERE user_id = %s", (user_id,))
result = cursor.fetchone() or {}

stored_skills = []
raw_skills = result.get("extracted_skills")
if raw_skills:
try:
stored_skills = json.loads(raw_skills)
except Exception:
stored_skills = [skill.strip() for skill in str(raw_skills).split(",") if skill.strip()]

resume_text = ""
resume_pdf = result.get("resume_pdf")
if resume_pdf:
resume_text = extract_text_from_pdf(BytesIO(resume_pdf))
return stored_skills, resume_text
finally:
cursor.close()
conn.close()
def render_ats_resume_analyzer():
# Detailed ATS analysis UI is intentionally disabled.
# Minimal score/progress/basic feedback is shown in the Update Resume flow.
# Keep this function to preserve compatibility with existing app flow.
return
```

```
try:
# Safely attempts to read from secrets
if st.secrets.get("MAINTENANCE_MODE") == "True":
st.error("🛠️ **System Offline for Maintenance**")
st.warning("SmartCampus app is being modified for your ease of use! Please check back after sometime. ")
st.stop()
except Exception:
# If the secrets file doesn't exist (like on your localhost), it just silently passes
pass
# Your normal app code continues below this...
def main():
st.title("Welcome to SmartCampus")

def main():
init_db()
init_admin_settings()
init_session_state()
if get_maintenance_mode():
if st.session_state.get('user_role') != 'admin':
st.empty()
st.title("🚧 Site Under Maintenance")
st.warning("SmartCampus is currently undergoing scheduled IT maintenance. We'll be back online shortly.")
st.info("Your application progress and profile data are safely stored in our database.")
st.divider()

with st.expander("Super Admin Override"):
st.caption("Only accounts with the 'admin' role can unlock the platform.")
admin_email = st.text_input("Admin Email", key="maint_em")
admin_pw = st.text_input("Admin Password", type="password", key="maint_pw")

if st.button("Bypass Lock", type="primary"):
user = authenticate_user(admin_email, admin_pw)
if user and user['role'] == 'admin':
st.session_state.logged_in = True
st.session_state.user_role = user['role']
st.session_state.user_id = user['id']
st.session_state.user_name = user['name']
new_token = str(uuid.uuid4())
register_new_session(user['id'], new_token)
st.session_state.session_token = new_token
```

```
st.success("Admin access granted.")
time.sleep(1)
st.rerun()
else:
st.error("Access Denied. Invalid credentials or insufficient permissions.")

st.stop()
if not st.session_state.logged_in:
p_home = st.Page(show_home, title="Welcome", icon="🏠", default=True)
p_login = st.Page(show_login, title="Login", icon="🔑")
p_register = st.Page(show_register, title="Register", icon="📝")

if 'pages' not in st.session_state:
st.session_state.pages = {
'home': p_home,
'login': p_login,
'register': p_register
}
pg = st.navigation([p_home, p_login, p_register], position="hidden")
pg.run()
else:
role = st.session_state.user_role

if role == 'student':
pg = st.navigation([st.Page(student_dash, title="Student Dashboard",
icon="➳")], position="hidden")
elif role == 'coordinator':
pg = st.navigation([st.Page(coordinator_dash, title="Coordinator Dashboard",
    icon="👕")], position="hidden")
elif role == 'admin':
pg = st.navigation([st.Page(admin_dash, title="Admin Control Center",
icon="♡□")], position="hidden")

pg.run()
st.divider()
if st.button("🚪 Secure Logout", type="primary"):
destroy_session(st.session_state.get('session_token'))
try:
controller.remove('smartcampus_user')
except Exception:
pass
```

```
st.session_state.clear()
st.rerun()
if __name__== "__main__":
main()

import random
from database import reset_password, delete_user, get_registration_status,
    set_registration_status, get_connection, \
get_engine_status, set_engine_status

def init_admin_settings():
conn = get_connection()
if not conn: return
cursor = conn.cursor()
try:
cursor.execute("""
CREATE TABLE IF NOT EXISTS system_settings (
setting_key VARCHAR(50) PRIMARY KEY,
setting_value VARCHAR(50) NOT NULL
)
""")
cursor.execute(
"INSERT IGNORE INTO system_settings (setting_key, setting_value)
VALUES ('maintenance_mode', '0')")
conn.commit()
except Exception as e:
print(f"DB Init Error: {e}")
finally:
cursor.close()
conn.close()

def get_maintenance_mode():
conn = get_connection()
if not conn: return False
cursor = conn.cursor()
try:
cursor.execute("SELECT setting_value FROM system_settings WHERE
setting_key = 'maintenance_mode'")
res = cursor.fetchone()
return res[0] == '1' if res else False
except:
return False
finally:
cursor.close()
```

```
conn.close()

def set_maintenance_mode(is_active):
conn = get_connection()
if not conn: return False
cursor = conn.cursor()
val = '1' if is_active else '0'
try:
cursor.execute("UPDATE system_settings SET setting_value = %s WHERE
setting_key = 'maintenance_mode'", (val,))
conn.commit()
return True
except:
return False
finally:
cursor.close()
conn.close()

def run_cli_toolkit():
print("\n=== SmartCampus Admin Toolkit ===")
print("1. Force Password Reset (Auto-Generate PIN)")
print("2. Delete a user account")
print("3. Toggle New Account Registrations")
print("4. Toggle Maintenance Mode")
print("5. Toggle AI Engines (ML / LLM / API)")

choice = input("\nSelect an option (1, 2, 3, 4, or 5): ")

if choice == '1':
target_email = input("Enter the user's email address: ")
temp_pin = str(random.randint(100000, 999999))
success, message = reset_password(target_email, temp_pin

if success:
print(f"\n✅ SUCCESS: Account Locked & Reset!")
print(f"🔋 TEMPORARY PIN: {temp_pin}")
print("The user MUST change this on their next login.")
else:
print(f"\n❌ FAILED: {message}")

elif choice == '2':
target_email = input("Enter the email of the account to DELETE: ")
confirm = input(f"Are you sure you want to delete {target_email}? (y/n): ")
if confirm.lower() == 'y':
```

```
success, message = delete_user(target_email)
print(f"\nResult: {message}")
else:
print("\nDeletion cancelled.")

elif choice == '3':
current_status = get_registration_status()
print(f"\nCurrent Registration Status: {'OPEN' if current_status else 'CLOSED'}")
print("1. OPEN Registrations")
print("2. CLOSE Registrations")
toggle_choice = input("Select an option (1 or 2): ")
set_registration_status(True if toggle_choice == '1' else False)
print("\nResult: Registration status updated.")
elif choice == '4':
current_maint = get_maintenance_mode()
print(f"\nCurrent Maintenance Status: {'ACTIVE' if current_maint else 'OFF'}")
print("1. ACTIVATE Maintenance Mode (Lock site)")
print("2. DEACTIVATE Maintenance Mode (Unlock site)")
toggle_choice = input("Select an option (1 or 2): ")
set_maintenance_mode(True if toggle_choice == '1' else False)
print("\nResult: Maintenance mode updated.")
elif choice == '5':
print("\n--- AI Engine Status ---")
print(f"ML Engine:  {'ON' if get_engine_status('ml_engine') else 'OFF'}")
print(f"LLM Engine: {'ON' if get_engine_status('llm_engine') else 'OFF'}")

print(f"API Engine: {'ON' if get_engine_status('api_engine') else 'OFF'}")

print("\n1. Toggle ML Engine")
print("2. Toggle LLM Engine")
print("3. Toggle API Engine")
eng_choice = input("Select an option (1, 2, or 3): ")
if eng_choice == '1':
curr = get_engine_status('ml_engine')
set_engine_status('ml_engine', not curr)
print(f"\nResult: ML Engine is now {'OFF' if curr else 'ON'}")
elif eng_choice == '2':
curr = get_engine_status('llm_engine')
print(f"\nResult: API Engine is now {'OFF' if curr else 'ON'}")
if __name__== "__main__":
init_admin_settings()
run_cli_toolkit()
```

OUTPUT

1. Main Portal

Fig A. Centralized access and user login

2. AI Resume Parsing & Student Profile

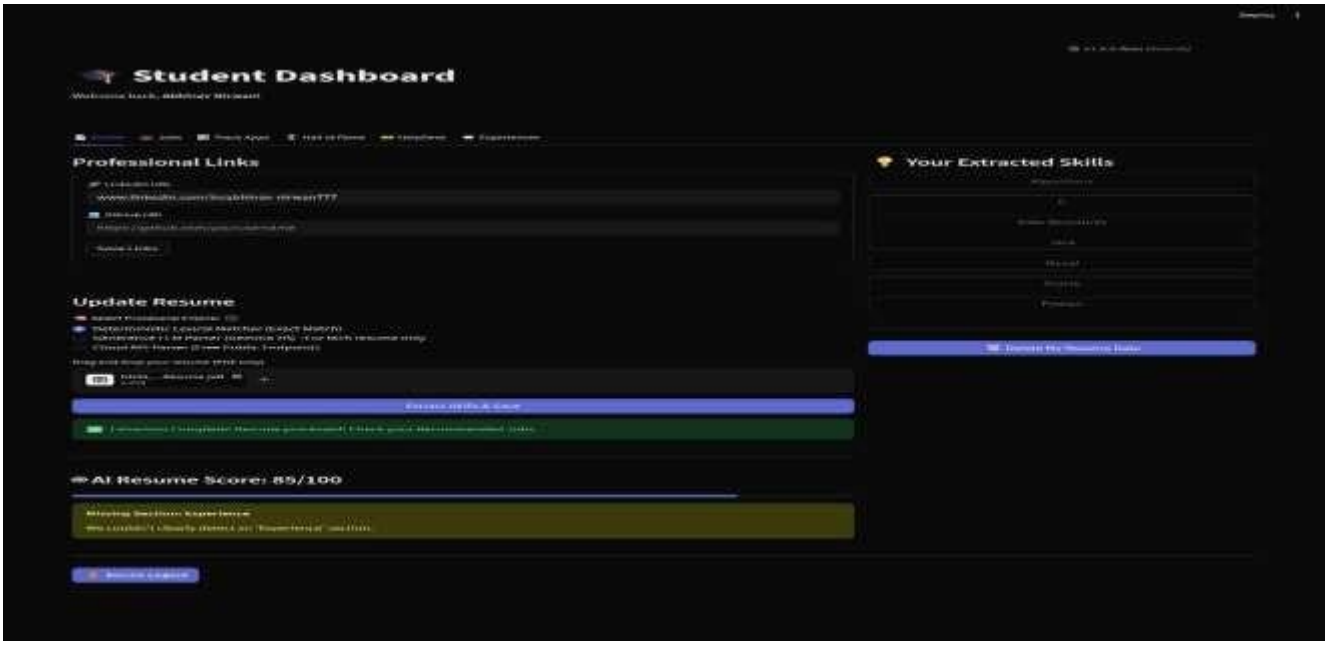

Fig B. AI Resume & Readiness Scoring

3. Smart Job Matching & Analytics

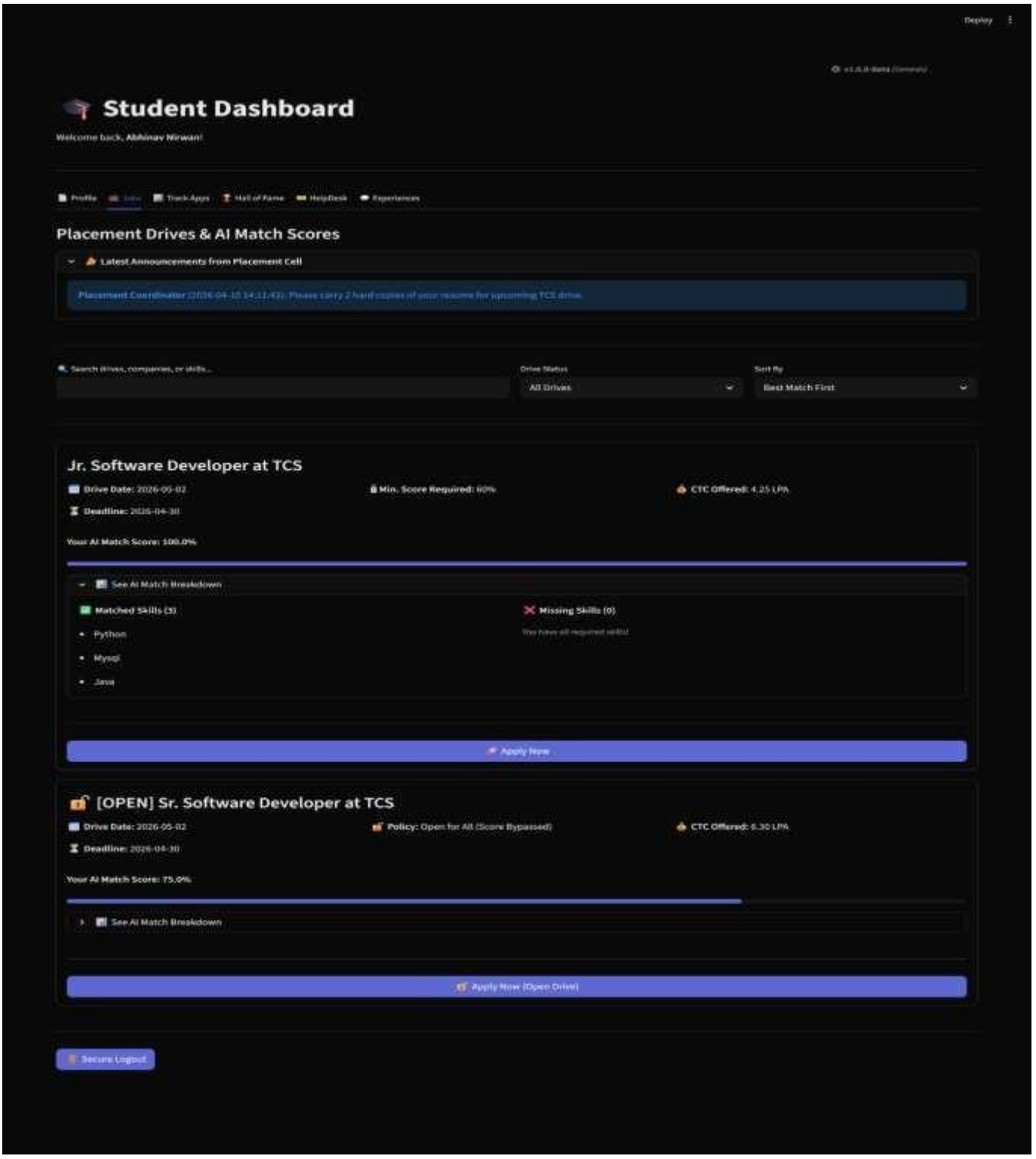

Fig C. Job recommendations with skill insights.

4. Placement Cell Controls & Admin Settings

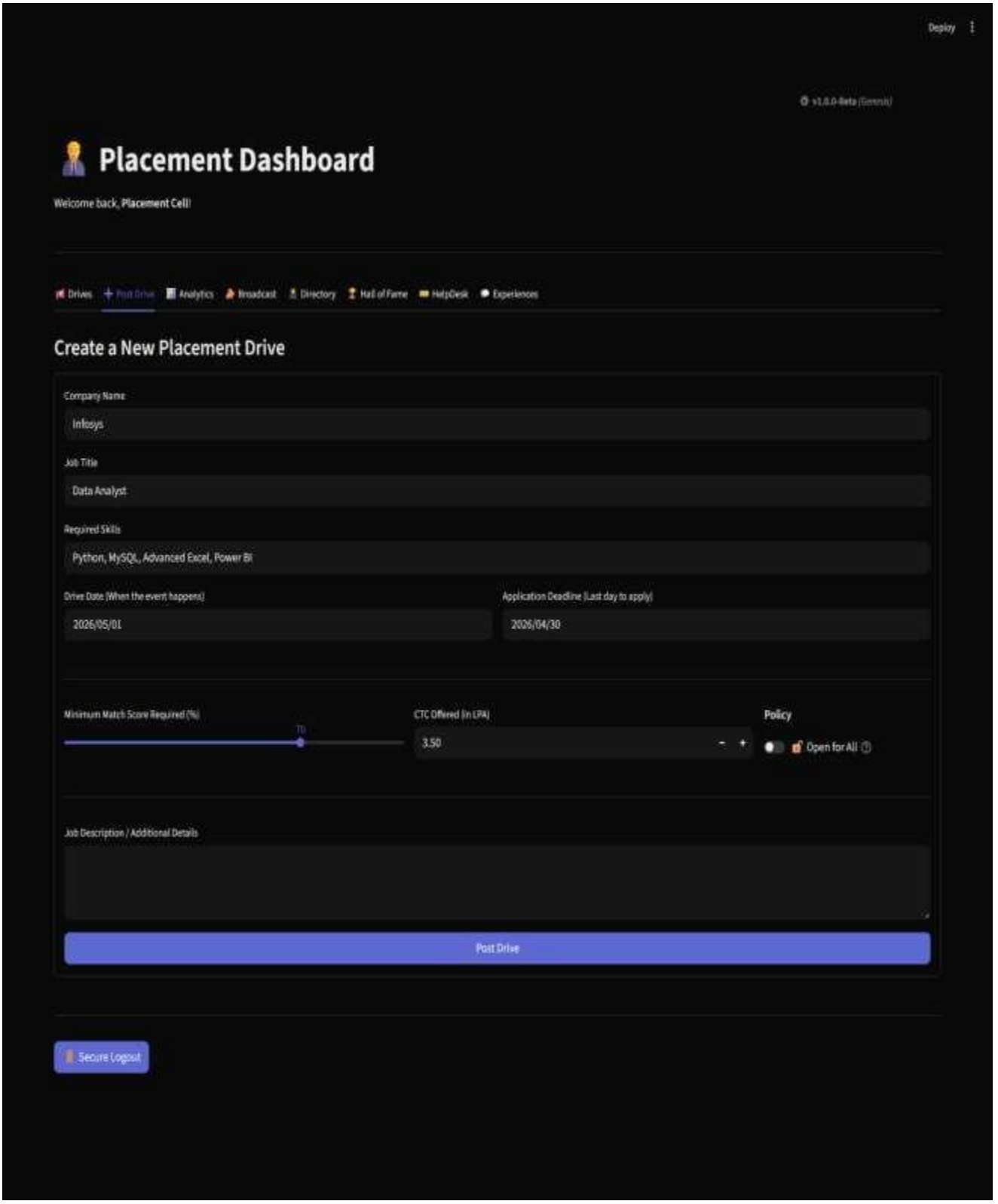

Fig D. Placement Drive Management

5. Admin control center

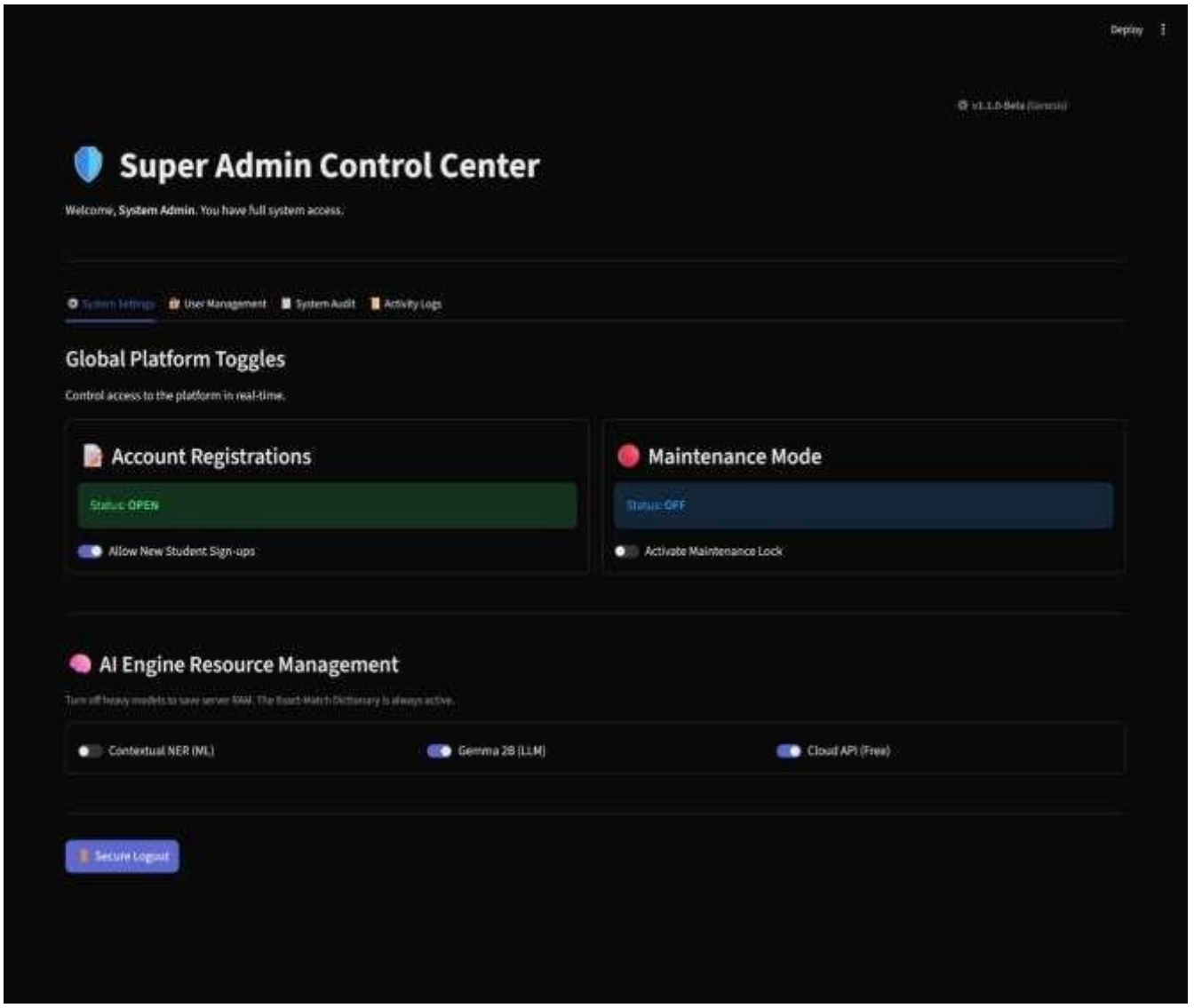

Fig E. System Settings & Performance

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